

Controller design

$$\mathcal{S}(a(s), b(s), m+1) \begin{pmatrix} p_0 \\ \vdots \\ p_m \\ q_0 \\ \vdots \\ q_m \end{pmatrix} = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n+m} \end{pmatrix}$$

$$C(s) = \frac{q_0 s^m + \dots + q_m}{p_0 s + \dots + p_m}$$

Strictly controller design

$$\begin{cases} p_0 = b/a_0 \\ \mathcal{S}(a(s), b(s)) \begin{pmatrix} p_1 \\ \vdots \\ p_m \\ q_1 \\ \vdots \\ q_m \end{pmatrix} = \begin{pmatrix} c_1 - a_1 c_0 / a_0 \\ \vdots \\ c_n - a_n c_0 / a_0 \\ c_{n+1} \\ \vdots \\ c_{n+m} \end{pmatrix} \end{cases}$$

Second-order system

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \sigma)^2 + \omega_d^2}$$

$$\omega_n = \sqrt{a_0}, \quad \zeta = \frac{a_1}{2\omega_n} = \frac{a_1}{2\sqrt{a_0}}$$

$$s = -\omega_n \zeta \pm \omega_n \sqrt{\zeta^2 - 1} = -\sigma \pm j\omega_d$$

$$\sigma = \zeta\omega_n, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$\zeta \leq 0 \Rightarrow$ unstable system

$|\zeta| < 1 \Rightarrow$ oscillatory system

$\zeta > 1 \Rightarrow$ over damped

$\zeta \approx 1 \Rightarrow$ critically damped

$\zeta < 0 \Rightarrow$ under damped

$$t_r = \frac{1.8}{\omega_n}$$

$$t_p = \frac{\pi}{\omega_d}$$

$$M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

$$t_s = \frac{3}{\sigma}$$

Root locus method : $1 + K \frac{b(s)}{a(s)}$

Rule A: # of branches = $\deg(a(s))$

Rule B: Starting points : $a(s) = 0$

Rule C: Ending points : $b(s) = 0$

Rule D:

Rule E: $\theta = \frac{(2l+1) \times 180^\circ}{m-n}$ $l = \{0, 1, \dots, m-n-1\}$

Rule F: 1. $a + Kb$, find K for stability

2. Solve for $a(j\omega) + K_{critical} b(j\omega) = 0$